

My QUESTIONS

MATH CLUB COORDINATOR TENURE

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Concept Map of Questions

1. Optimal Prefix Coding (Huffman Encoding)
 2. Binary Information Capacity & Encoding
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§1 Coord App

1. Consider a hypothetical world M, where people speak in a language which has only A, B, C, D, E characters to use (no space as well). The dogs in this world are very Enthu dogs, who want to learn their own language to communicate what people are speaking among themselves. But, the dogs can only bark or not bark.

So, all the dogs came up with a strategy to translate the human language to their bark language.

Say, ABE in human language is (nnnnnbbnn) in the bark language where n stands for *no bark* and b stands for *bark*.

The intelligent dogs observed the frequency of letters in the human language and realised that they could change their strategy of translation so as to reduce the number of b and n in their language and still convey the whole human message correctly. This is the most optimal bark language. Also, generally, the number of b is less than the number of n .

Find out the most optimal bark language message of the human language message

ABEDCBADADAABCDABEBCABCDABECAAABBABACEDABBAAABCEAABAAB

Assume that the frequency of letters in human language always remains same.

Solution:

This is based on the [Huffman Encoding Algorithm](#).

§2 Mock Interns

2. You are given 1000 unmarked bottles of juice. They are identical in appearance, smell, and taste—but exactly one is laced with a deadly poison. The poison has a unique property: it causes death exactly 24 hours after consumption. You also have access to 10 perfectly healthy prisoners, and you may use them to test the bottles in any way you like—but you are allowed only one round of testing, and you must identify the poisoned bottle within 24 hours.

Fortunately, you're clever. You devise a strategy that ensures you can pinpoint the exact poisoned bottle using at most 10 prisoners, and you do so successfully with the 1000 bottles.

Now, using the same strategy, under the same rules and constraints, what is the largest number of bottles you could have handled and still guaranteed identifying the poisoned one within 24 hours?

Solution:

Core Insight: Each prisoner gives us 1 bit of information: they are either dead (1) or alive (0) after 24 hours. So, with 10 prisoners, we get 10 bits of total information from a single round of testing.

Since 10 bits can represent up to $2^{10} = 1024$ distinct binary combinations, this means we can uniquely identify up to 1024 different bottles (i.e., poisoned bottle positions) using 10 binary outcomes.

But the question mentions only 1000 bottles, so we are not fully using the capacity of our 10-bit system.

So, the **strategy** that you device is:

- Label each bottle with natural numbers.

- Convert each bottle number into its 10-bit binary representation (e.g., bottle #25 = 0000011001)
- Assign each prisoner a bit position from 1 to 10 (e.g., prisoner 1 checks bit 1 (LSB), prisoner 10 checks bit 10 (MSB)).
- For each bottle: If a bottle's binary representation has a 1 in the position assigned to a prisoner, that prisoner drinks a drop of juice from that bottle.
- After 24 hours, some prisoners die.
- The combination of dead/alive prisoners gives a 10-bit binary number.
- Convert that binary number back to decimal: this number is the poisoned bottle.

So, this strategy can be extended upto 1024 bottles, which is the capacity of the 10-bit number system.

Thus, the answer is 1024.

3. You are given a set $M \subset \{1, 2, 3, \dots, 64\}$ such that $|M| = 10$. Let A and B be two **non-empty, disjoint subsets** of M having the **same sum** of elements.

Which of the following is true?

- Mark **1** if $\exists A, B$ for **every** such set M
- Mark **2** if $\exists A, B$ for **only some** such sets M
- Mark **3** if $\nexists A, B$ for **any** such set M

Solution:

For any 10-element subset $M \subset \{1, 2, \dots, 64\}$:

- The number of **non-empty, proper subsets** of M is $2^{10} - 2 = 1022$.
- The **maximum possible subset sum** occurs when $M = \{55, 56, \dots, 64\}$, yielding:

$$\sum_{i=56}^{64} i = 540.$$

- Therefore, the number of **distinct possible subset sums** is at most 540.
- By the **Pigeonhole Principle**, at least two distinct subsets of M must have the same sum.
- Even if these two subsets are not disjoint, we can always construct two disjoint subsets with the same sum from them.

Conclusion: For **every** 10-element subset M , there exist two **non-empty, disjoint subsets** A and B with equal sums.

Hence, **Answer** = 1

4. Consider $E(x,y) = 0$ and $H(x,y) = 0$ be the equations of **some** Ellipse and Hyperbola respectively. Let, $S(x,y)$ be a second-degree polynomial.

It is known that, $\exists a, b \in \mathbb{R}$:

$$E(x^2, xy) + 3 \times H(xy, y^2) = xy \times S(x, y) + ax^4 + by^4$$

Now,

$$\frac{\partial S(x, y)}{\partial x} = 0 \text{ and } \frac{\partial S(x, y)}{\partial y} = 0 \text{ pass through a point } (\alpha, \beta) \quad (1)$$

Find $\alpha + \beta$.

Solution:

Here, there is very little information given about the polynomials. So, we can use the fact that **E(x,y), H(x,y) and S(x,y) are polynomials** and that the **degree of each term on LHS = RHS**.

Now, the general second degree equation is $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$. $E(x^2, xy)$ will contain the terms $x^4, x^2y^2, x^3y, x^2, xy, c$ and $H(xy, y^2)$ will contain $x^2y^2, y^4, xy^3, xy, y^2, c$.

The RHS contains the terms $x^3y, xy^3, x^2y^2, x^2y, xy^2, xy, x^4, y^4$

Now, we see that LHS has the terms x^2, y^2 extra. But, these can be safely assumed to be zero as they are not essential terms for an Ellipse and Hyperbola. [Also, it is not said that this equation is true for every ellipse and every hyperbola].

The RHS has x^2y and xy^2 terms as extra which came from the $2gx$ and $2fy$ terms of $S(x,y)$. So, for this equation to hold, we should have $2g = 0$ and $2f = 0$ of $S(x,y)$.

Consider the general equation of such an $S(x,y)$ to be $ax^2 + by^2 + 2hxy + c$.

Now,

$$\frac{\partial S(x, y)}{\partial x} = 2ax + 2hy \text{ and } \frac{\partial S(x, y)}{\partial y} = 2by + 2hx \text{ pass through } (0,0) \quad (2)$$

Hence, $\alpha = 0$ and $\beta = 0$. **Answer = 0**

5. Consider a multi-variable function $f : \mathbb{R}^9 \rightarrow \mathbb{R}$. Let the input variables be $a_i, i \in \{1, 2, 3, \dots, 9\}$. The following facts about the function is known

- $f(a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9) = -f(a_4, a_5, a_6, a_1, a_2, a_3, a_7, a_8, a_9)$
- $f(1, 0, 0, 0, 1, 0, 0, 0, 1) = 1$
- $f(a_1, a_2, a_3, a_7, a_8, a_9, a_4, a_5, a_6) = -f(a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9)$
- $f(a_1, a_2, a_3, ta_4, ta_5, ta_6, a_7, a_8, a_9) = t \times f(a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9) \quad \forall t \in \mathbb{R}$
- $f(a_7, a_8, a_9, a_4, a_5, a_6, a_1, a_2, a_3) = -f(a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9)$
- $f(pa_1, pa_2, pa_3, a_4, a_5, a_6, a_7, a_8, a_9) = p \times f(a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9) \quad \forall p \in \mathbb{R}$
- $f(a_1, a_2, a_3, a_4, a_5, a_6, a_7 + a'_7, a_8 + a'_8, a_9 + a'_9) =$
 $f(a_1, a_2, a_3, a_4, a_5, a_6, a_7, a'_8, a'_9) + f(a_1, a_2, a_3, a_4, a_5, a_6, a'_7, a_8, a_9)$
- $f(a_1, a_2, a_3, a_4, a_5, a_6, ra_7, ra_8, ra_9) = r \times f(a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9) \quad \forall r \in \mathbb{R}$
- $f(a_1, a_2, a_3, a_4 + a'_4, a_5 + a'_5, a_6 + a'_6, a_7, a_8, a_9) =$
 $f(a_1, a_2, a_3, a'_4, a_5, a'_6, a_7, a_8, a_9) + f(a_1, a_2, a_3, a_4, a'_5, a_6, a_7, a_8, a_9)$
- $f(a_1 + a'_1, a'_2 + a_2, a'_3 + a_3, a_4, a_5, a_6, a_7, a_8, a_9) =$
 $f(a'_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9) + f(a_1, a'_2, a'_3, a_4, a_5, a_6, a_7, a_8, a_9)$

Now, find the value of $f(5, 1, 3, 8, -4, 2, 1, 0, -3)$

Solution:

Here, the key thing to identify is that the properties described here are the defining properties of the determinant of a 3x3 matrix.

$$f(a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9) = \begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$$

Also, since these are the defining properties of a determinant, we can prove all the other properties of determinant and hence derive the formula for the determinant. Hence,

$$f(5, 1, 3, 8, -4, 2, 1, 0, -3) = \begin{vmatrix} 5 & 1 & 3 \\ 8 & -4 & 2 \\ 1 & 0 & -3 \end{vmatrix} = 98$$

Therefore, the **answer is** 98

6. Consider an extended Chess game on an $n \times n$ board. Let, $h(n)$ denote the number of board positions which consists of only n non-attacking rooks [no two rooks should attack each other] in such a way that none of them lie on the main diagonal from bottom left to top right corner [only one of the diagonal is forbidden not both]. The rooks behave normally as in Chess. Find the GIF of

$$h(5) + e^2 \times \left(\lim_{n \rightarrow \infty} \frac{h(n)}{n!} \right)$$

* GIF - Greatest Integer Function

Solution:

Rook covers a row and a column. And since the number of rooks = number of rows = number of columns = n , each row and each column should have one and only one rook for them to be non-attacking. Now, if there was no constrain on the main diagonal, it would have been just the permutations along either rows or columns = $n!$. But, the constraint that the rooks should not be on the main diagonal suggests us about Derangement. So, $h(n)$ = number of ways in which row i can be mapped to a column other than $i \forall i < n$. So, say if row i is mapped to col j , $i \neq j$, then the rook must be placed in row i and col j . So, $h(n)$ = Derangement(n) So,

$$h(n) = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} \dots + \frac{(-1)^n}{n!} \right)$$

Now, $h(5) = 44$ Also,

$$\lim_{n \rightarrow \infty} \frac{h(n)}{n!} = \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} \dots \right) = \frac{1}{e}$$

So, the **answer** = $[44 + e] = [46.71] =$ 47

§3 Freshie Roadmaps

7. Consider a set of 3 cities in India. The number of roads from one city to another is given in the table below. All the roads are one-way and of unit length. In this matrix, $a_{i,j}$ is the number of roads from city i to city j . Also, the diagonal entries represent the roads that begin and end in the same city. Such a matrix is known as an **adjacency matrix** which stores the number of ways of transition from one state to another. A **closed loop** is defined as a path that starts and ends at the same city. Let $f(n)$ denote the total number of different closed loops of length n from all the cities.

$$A = \begin{bmatrix} 3 & 15 & 6 \\ 1 & 5 & 2 \\ 2 & 10 & 4 \end{bmatrix}$$

Now, consider the following Python program:

```

1  sum = 0
2  count = 1
3
4  for i in range(1,n):
5      count *= i
6      sum += f(i) / count
7
8  return sum

```

What is the output of this program as $n \rightarrow \infty$?

Solution:

Number of ways to go from city i to j after n transitions is the (i, j) th element of A^n . So, $f(n) = \text{trace}(A^n)$

$\text{count} = i!$

So, the answer is $\sum_{i=1}^{\infty} \frac{\text{trace}(A^i)}{i!}$

Observe that $A^2 = 12A$. So, $A^n = 12^{n-1}A \Rightarrow \text{trace}(A^i) = 12^{i-1} \times \text{trace}(A) = 12^{i-1} \times 12 = 12^i$

Hence, answer = $\boxed{e^{12} - 1}$

§4 DC App

8. **Infinite Candy Store:** There is a magical candy store that operates under a curious rule. Whenever you visit the store and give them x chocolates and y toffees, they return to you

$$(\text{chocolates}, \text{toffees}) = (x + 3y, 8x + 3y).$$

You are allowed to visit the store only once per day. Each day, you hand over *all* the chocolates and toffees you have/received from the previous day to get new ones in return. You decided that you will eat everything only after n days when you will have exactly 2401 chocolates and 4802 toffees. Assume that the first time you visited the store (day 0), you had just one chocolate and t_0 toffees.

Problem 3.2.1: Find t_0 so that you'll eat 2401 chocolates and 4802 toffees some day in future.

Problem 3.2.2: Find $\min(n)$

(Brownie Question) Prove that this is the only solution (t_0, n) possible.

Solution:

This is where the beauty of Linal shines. You can write the number of chocolates and toffees that you'll have after n days in this compact and useful way

$$\begin{bmatrix} c_n \\ t_n \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 8 & 3 \end{bmatrix}^n \begin{bmatrix} c_0 \\ t_0 \end{bmatrix}$$

This can be re-written as

$$\begin{bmatrix} 2401 \\ 4802 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 8 & 3 \end{bmatrix}^n \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

As a rule of thumb, when ever you see a matrixⁿ, find it's eigen values and eigen vectors. Explain why this is useful in the interview.

The eigenvalues and corresponding eigenvectors of A are:

$$\lambda_1 = 7, \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{and} \quad \lambda_2 = -3, \quad \mathbf{v}_2 = \begin{bmatrix} -3 \\ 4 \end{bmatrix}.$$

Now, if we consider the case of $(c_n, t_n) = (1, 2)$, then the equation becomes $7^n = 2401 \Rightarrow \boxed{n = 4}$.

This gives us $\boxed{t_n = 2}$.

We can prove that this is the only solution as $(c_n, t_n) = (2401, 4802)$ is also an eigen vector of this matrix. So, since this matrix is invertible, (c_{n-1}, t_{n-1}) should also be an eigen vector, and by induction, we get (c_0, t_0) must be an eigen vector.

Geometrically, eigen vectors after transformation lie along the same line.

§5 Advent Calendar

9. Consider an equilateral triangular wire frame of side length 10. The sides have a uniform linear charge density $\lambda_1 = 2, \lambda_2 = 3$ and $\lambda_3 = 5$ respectively. P is the point where the net potential due to these sides is maximum. Find the value of $\frac{1}{2k} \times (|\mathbf{E}_1| + |\mathbf{E}_2| + |\mathbf{E}_3|)$ at point P, where $|\mathbf{E}_i|$ is the magnitude of electric field due to λ_i .

Assume that $V = 2k\lambda \ln(r)$ and $|\mathbf{E}| = 2k\frac{\lambda}{r}$ is the potential and electric field due λ at distance r . Consider all quantities(including the final answer) in SI units.

Solution:

Let r_1, r_2, r_3 be the perpendicular distances from the sides of the triangle.

At point P, $V = 2k \times (\lambda_1 \ln r_1 + \lambda_2 \ln r_2 + \lambda_3 \ln r_3)$ is maximum.

Now, since r_1, r_2, r_3 are distances inside a finite triangle, they must be bounded and hence V has a

finite upper bound.

Since it's an equilateral triangle, [Viviani's Theorem*](#) gives us $r_1 + r_2 + r_3 = 10 \cdot \frac{\sqrt{3}}{2} = 5\sqrt{3}$

Now, it's like we have to maximise a multivariable function given some constraints. This is exactly what [Lagrange multiplier](#) method is for.

Maximise $V = 2k \times (\lambda_1 \ln r_1 + \lambda_2 \ln r_2 + \lambda_3 \ln r_3)$ given $g(r_1, r_2, r_3) = r_1 + r_2 + r_3 - 5\sqrt{3} = 0$

Since the maximum occurs when

$$\nabla V = \mu \nabla g$$

for some constant μ , we compute the partial derivatives.

$$\frac{\partial V}{\partial r_i} = 2k \cdot \frac{\lambda_i}{r_i}, \quad i = 1, 2, 3$$

and

$$\frac{\partial g}{\partial r_i} = 1, \quad i = 1, 2, 3.$$

Thus, the Lagrange multiplier condition gives

$$\frac{2k\lambda_1}{r_1} = \frac{2k\lambda_2}{r_2} = \frac{2k\lambda_3}{r_3} = \mu.$$

Cancelling $2k$, we obtain

$$\frac{\lambda_1}{r_1} = \frac{\lambda_2}{r_2} = \frac{\lambda_3}{r_3}.$$

Hence,

$$r_1 : r_2 : r_3 = \lambda_1 : \lambda_2 : \lambda_3.$$

Using the constraint

$$r_1 + r_2 + r_3 = 5\sqrt{3},$$

we get

$$r_i = \frac{5\sqrt{3}\lambda_i}{\lambda_1 + \lambda_2 + \lambda_3}, \quad i = 1, 2, 3.$$

$$\text{Now, } \frac{1}{2k} \times (|\mathbf{E}_1| + |\mathbf{E}_2| + |\mathbf{E}_3|) = \frac{\lambda_1}{r_1} + \frac{\lambda_2}{r_2} + \frac{\lambda_3}{r_3} = \frac{\lambda_1 + \lambda_2 + \lambda_3}{5\sqrt{3}} \times 3 = \boxed{2\sqrt{3} = 3.464...}$$

*There's a really nice visual proof of [Viviani's Theorem here](#).

10. Let $S_b(n)$ be the sum of digits of n when expressed in base b .

$$f_n = S_7(S_7(S_7(\dots S_7(13^{45} + 14^{46} + 15^{47})\dots)))$$

where the function $S_7()$ is applied n times. Find the limiting value of the sequence

$$f_1, f_2, f_3, \dots$$

Solution:

$S_b(n)$ is the digit sum of n in base b .

Let

$$n = a_k b^k + a_{k-1} b^{k-1} + \cdots + a_1 b + a_0$$

be the base- b representation of n , where $0 \leq a_i < b$.

Since

$$b \equiv 1 \pmod{b-1},$$

we have

$$b^k \equiv 1 \pmod{b-1}.$$

Therefore,

$$n \equiv a_k + a_{k-1} + \cdots + a_0 = S_b(n) \pmod{b-1}.$$

Applying this repeatedly shows that every iteration of S_b preserves the value modulo $b-1$. Hence the iterated digit sum converges to the digital root of n in base b , which is the unique integer $d \in \{1, 2, \dots, b-1\}$ such that

$$d \equiv n \pmod{b-1},$$

unless $n \equiv 0 \pmod{b-1}$, in which case the digital root is $b-1$.

Thus,

$$\lim_{n \rightarrow \infty} f_n = \text{digital root}_7(13^{45} + 14^{46} + 15^{47}).$$

Now compute modulo 6 (since $7-1=6$):

$$13 \equiv 1 \pmod{6}, \quad 14 \equiv 2 \pmod{6}, \quad 15 \equiv 3 \pmod{6}.$$

Hence,

$$13^{45} \equiv 1^{45} \equiv 1 \pmod{6},$$

$$14^{46} \equiv 2^{46} \equiv (2^2)^{23} \equiv 4^{23} \equiv 4 \pmod{6},$$

$$15^{47} \equiv 3^{47} \equiv 3 \pmod{6}.$$

since $4^k \equiv 4 \pmod{6}$ and $3^k \equiv 3 \pmod{6}$ for all $k \geq 1$

Adding,

$$13^{45} + 14^{46} + 15^{47} \equiv 1 + 4 + 3 = 8 \equiv 2 \pmod{6}.$$

Therefore, the digital root in base 7 is 2, and

$$\boxed{\lim_{n \rightarrow \infty} f_n = 2}.$$

11. Consider n vectors in $\mathbb{R}^n$ given by $v_i = \left[\frac{1}{6}, \frac{1}{6}, \dots, \frac{1}{2}, \dots, \frac{1}{6}, \frac{1}{6} \right]^T$ with $\frac{1}{2}$ in the i^{th} dimension and $\frac{1}{6}$ in all other dimensions.

Let, $V(n)$ = volume of the n -dim hyperparallelepiped formed by $v_1, v_2, \dots, v_n$. Find the sum of all such volumes for $n \in \mathbb{N}$, i.e., $\sum_{k=1}^{\infty} V(k)$

Solution:

The volume $V(n)$ is the absolute value of the determinant of the matrix whose columns are $v_1, v_2, \dots, v_n$.
Let, $V(n) = \det(A)$:

$$A = \begin{pmatrix} \frac{1}{2} & \frac{1}{6} & \cdots & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{2} & \cdots & \frac{1}{6} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{6} & \frac{1}{6} & \cdots & \frac{1}{2} \end{pmatrix}_{(n \times n)}$$

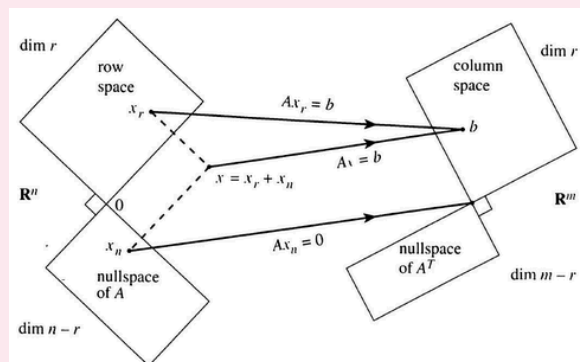
Since, n is a variable, doing row operations to find the $\det(A)$ is tough. So, if we are able to find all the eigen values of A , then we can find $\det(A)$ by multiplying all of them.

Observe the structure of this matrix A and $A - \frac{1}{3}I$.

$$\left| A - \frac{1}{3}I \right| = \begin{vmatrix} \frac{1}{6} & \frac{1}{6} & \cdots & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \cdots & \frac{1}{6} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{6} & \frac{1}{6} & \cdots & \frac{1}{6} \end{vmatrix}_{(n \times n)} = 0$$

This means that $\frac{1}{3}$ is an eigen value of A .

Now, $\left(A - \frac{1}{3}I \right)$ has rank 1. So, row-space of $\left(A - \frac{1}{3}I \right)$ has dimension 1 and null-space of $\left(A - \frac{1}{3}I \right)$ has dimension $(n - 1)$.



So, $\left(A - \frac{1}{3}I \right) x = 0$ has $(n - 1)$ independent solutions $\implies A$ has $(n - 1)$ independent eigen vectors $\implies$ geometric multiplicity of $\frac{1}{3}$ is $(n - 1)$.

Of the n eigen values, we now know that $(n - 1)$ of them are $\frac{1}{3}$.

To find the other eigen value, we know that the sum of all the eigen values = trace of the matrix.

So,

$$\frac{(n-1)}{3} + \lambda = \frac{n}{2} \implies \lambda = \frac{(n+2)}{6}$$

Since we know all the eigen values of A, the determinant of A = product of all the eigen values.

$$V(n) = \left(\frac{1}{3}\right)^{n-1} \times \frac{(n+2)}{6}$$

Now, $\sum_{k=1}^{\infty} V(k) = \sum_{k=1}^{\infty} \left(\frac{1}{3}\right)^{k-1} \times \frac{(k+2)}{6}$, which is an AGP, which turns out to be $\frac{7}{8}$

Answer = $\boxed{\frac{7}{8}}$

§6 Bids and Brains

12. Let

$$A = \int_0^1 \sin x [1 - (1-x)^5] dx$$

and

$$B = \int_0^1 \sin x [1 - x^5] dx$$

Find the value of

$$\lim_{n \rightarrow \infty} \frac{A^n}{A^n + B^n}$$

Solution:

This is a very tough integral to solve analytically. Observe that we need not find the value of A and B individually to answer this. We only need to find whether $A > B$ or $A < B$.

Now, observe that $\sin x$ is increasing in $[0, 1]$.

Also, $[1 - x^5]$ is decreasing in $[0, 1]$. So, $[1 - (1-x)^5]$ is increasing in $[0, 1]$.

From the **continuous version of Chebyshev's sum inequality**, we have:

- If both f and g are non-increasing or non-decreasing functions,

$$\int_0^1 f(x)g(x)dx \geq \left(\int_0^1 f(x)dx\right) \times \left(\int_0^1 g(x)dx\right)$$

- If f is non-decreasing and g is non-increasing,

$$\int_0^1 f(x)g(x)dx \leq \left(\int_0^1 f(x)dx\right) \times \left(\int_0^1 g(x)dx\right)$$

Now, consider $f(x)$ as $\sin x$ and $g(x)$ as $1 - x^5 \implies f$ is increasing and g is decreasing

So,

$$B = \int_0^1 \sin x [1 - x^5] dx \leq \left(\int_0^1 \sin x dx\right) \times \left(\int_0^1 [1 - x^5] dx\right)$$

Now, consider $f(x)$ as $\sin x$ and $g(x)$ as $1 - (1 - x)^5 \implies f$ and g both are increasing

So,

$$A = \int_0^1 \sin x [1 - (1 - x)^5] dx \geq \left(\int_0^1 \sin x dx \right) \times \left(\int_0^1 [1 - (1 - x)^5] dx \right)$$

Since $\int_0^1 [1 - (1 - x)^5] dx = \int_0^1 [1 - x^5] dx$, we have,

$$\boxed{A \geq B}$$

Hence, **answer = 1**

13. Among all functions $y : [0, 1] \rightarrow \mathbb{R}^+$ such that

$$y(0) = y(1) = 0 \quad \text{and} \quad \int_0^1 (y'(x))^2 dx = 1,$$

$y_0(x)$ is the function which maximizes the functional

$$A[y] = \int_0^1 x y(x) dx$$

Find the value of $\left\lfloor 8 \times \int_0^1 y(x) dx \right\rfloor$ where $\lfloor \cdot \rfloor$ is the floor function.

Solution:

This is a problem on calculus of variations with constraint given.

So, we cannot directly apply Euler-Lagrange Equation on $A[y]$, instead introduce a **Lagrange multiplier** λ and consider

$$J[y] = \int_0^1 (xy - \lambda(y')^2) dx$$

Since, y_0 maximises $A[y]$ and also satisfies the constraint, it is a stationary point of $J[y]$

The Euler-Lagrange equation is

$$\frac{d}{dx} \left(\frac{\partial L}{\partial y'} \right) - \frac{\partial L}{\partial y} = 0, \quad L = xy - \lambda(y')^2$$

Computing derivatives,

$$-2\lambda y'' - x = 0 \quad \Rightarrow \quad y'' = -\frac{x}{2\lambda}$$

Integrating twice,

$$y(x) = -\frac{x^3}{12\lambda} + C_1x + C_2$$

Using $y(0) = y(1) = 0$ gives

$$y(x) = \frac{1}{12\lambda}(x - x^3)$$

Now enforce the constraint:

$$y'(x) = \frac{1}{12\lambda}(1 - 3x^2),$$

$$\int_0^1 (y')^2 dx = \frac{1}{144\lambda^2} \int_0^1 (1 - 3x^2)^2 dx = \frac{1}{180\lambda^2}$$

Setting this equal to 1 yields

$$\lambda = \frac{1}{6\sqrt{5}}$$

Therefore, the maximizing function is

$$y(x) = \frac{\sqrt{5}}{2}(x - x^3)$$

$$\text{Now, } 8 \times \int_0^1 y(x) dx = \sqrt{5} \implies \left[8 \times \int_0^1 y(x) dx \right] = 2$$

$$\text{Answer} = \boxed{2}$$

14. Consider the functional equation:

$$f(x + f(y)) = f(x) + y \quad \forall x, y \in \mathbb{R}$$

Find the number of real valued functions, $f : \mathbb{R} \rightarrow \mathbb{R}$ that satisfy this functional equation.

Solution:

Let's test for **Injectivity**:

Assume $f(a) = f(b)$: Put $y = a$ and $y = b$ in the functional equation:

$$f(x + f(a)) = f(x) + a \quad f(x + f(b)) = f(x) + b$$

From these, we get $a = b$

This proves that the function has to be injective (one-one).

Let's test for **Surjectivity**:

Put $x = 0$:

$$f(f(y)) = f(0) + y$$

Since, RHS is unbound and $f(0) + y \in \mathbb{R}$, we should have $f(\cdot)$ to be able to take any real value.

This proves that the function has to be surjective(onto).

From, both of these, we can say that the function has to be **Bijjective**

Now, $x \leftrightarrow y$:

$$f(y + f(x)) = f(y) + x \quad \forall x, y \in \mathbb{R}$$

Substituting this back in the given equation, we get:

$$f(f(f(x) + y)) = f(x) + y \quad \forall x, y \in \mathbb{R} \implies f(f(t)) = t \quad \forall t \in \mathbb{R}$$

Since, $f(x)$ is Bijjective, it is also invertible

So, $f(t) = f^{-1}(t) \quad \forall t \in \mathbb{R}$

This is possible only if $f(x) = \pm x \quad \forall x \in \mathbb{R}$

Answer = **2**

